SOFTWARE REQUIREMENTS SPECIFICATION

Hire for Help

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1. Introduction

1.1 Purpose

This SRS defines the minimum functional, data, interface, security, and operational requirements for a local marketplace that connects customers with service workers. It is intended to guide implementation and testing. Requirements marked as dependent on provider or legal confirmation must be resolved before production launch.

1.2 Product Scope

The platform enables customers to discover workers, request and manage bookings, receive platform-funded discounts, pay through an integrated payment provider, and view transaction history. Workers can register, submit verification information, publish skills and availability, manage job requests, and view settlement status. Administrators manage verification, bookings, promotions, disputes, and operational records.

2. Product Overview

2.1 Product Goals

- Make it easy for customers to find and book suitable local workers.
- Provide a clear, auditable booking and payment journey.
- Build trust through worker verification and reviews.
- Support platform-funded promotional discounts without reducing agreed worker earnings unless explicitly agreed.
- Launch in one locality and expand only after usage and operational performance are validated.

3. User Classes and Permissions

Role	Main capabilities	Restrictions
Customer	Verify mobile number; manage profile/addresses; search workers; book/cancel within policy; pay; view own transactions; review completed jobs; raise disputes.	May access only their own private data and bookings.
Worker	Manage profile, skills, service area, availability; submit verification; accept/reject eligible requests; update job status; view own settlements; respond to operational requests.	Cannot accept jobs until required verification and payout onboarding are approved.
Administrator	Review worker verification; manage accounts, service categories, promotions, bookings, refunds/disputes and operational reports.	Privileged actions must be permission-controlled and audit logged.
System / provider	Process payment callbacks, notifications, and scheduled status updates.	Only authenticated, validated provider events may update financial status.

4. Functional Requirements

Priority labels: P0 = required for MVP; P1 = important but may follow the first controlled pilot.

ID	Requirement	Priority	Specification
FR-01	Registration and authentication	P0	The system shall support customer and worker registration using a mobile number and OTP. OTPs must expire, be rate-limited, and never be stored in plaintext. Sessions shall be securely issued and invalidated.
FR-02	Role-based access	P0	The system shall enforce customer, worker, and administrator permissions on every protected API operation.
FR-03	Customer profile	P0	Customers shall manage basic profile details and service addresses.

FR-04	Worker profile	P0	Workers shall manage their name, description, skills, experience, service area, pricing, and availability.
FR-05	Worker verification	P0	Workers shall submit required verification details through an approved provider or documented manual review process. Verification status and review history shall be recorded. Unapproved workers shall not accept bookings.
FR-06	Service catalogue and discovery	P0	Customers shall browse active service categories and search/filter eligible workers by service, area, availability, and supported price criteria.
FR-07	Booking request	P0	Customers shall submit a request containing worker/service, location, requested date/time, duration or scope, and a price summary. The system shall store a snapshot of agreed price and terms.
FR-08	Booking lifecycle	P0	Customers and workers shall see booking status. Only valid, role-authorized state transitions shall be permitted and recorded in status history.
FR-09	Promotions	P0	The system shall validate promotion dates, eligibility, usage limits, discount caps, and campaign budget before applying a platform-funded discount.
FR-10	Payment initiation	P0	The system shall create payment orders through a selected provider and associate provider references with the booking. Payment confirmation shall rely on verified provider events or provider-side status verification, not solely on the browser redirect.
FR-11	Payment webhook	P0	The system shall verify webhook authenticity, process events idempotently, reject invalid events, and record provider event identifiers to prevent duplicate processing.
FR-12	Transaction ledger	P0	Customers and workers shall see only their authorized financial history, including relevant payments, discounts, refunds, fees, and settlement statuses. The ledger shall not represent withdrawable or transferable stored funds.
FR-13	Worker settlement tracking	P0	The system shall record provider settlement references, amounts, status, and timestamps. Actual settlement shall use a provider-supported marketplace/payout workflow.
FR-14	Cancellation and refunds	P0	The system shall apply published cancellation rules and record refund requests, provider refund references, statuses, and final outcomes.
FR-15	Notifications	P0	The system shall notify relevant parties of key events such as booking request, acceptance/rejection, cancellation, payment outcome, and status changes through configured channels.
FR-16	Reviews	P1	Customers shall be able to submit a rating and optional review only for eligible completed bookings. Duplicate reviews shall be prevented.
FR-17	Disputes and support	P0	Customers and workers shall be able to raise a dispute or support issue linked to a booking. Administrators shall record status, decision, and resolution notes.
FR-18	Administration and audit	P0	Administrators shall manage service categories, worker verification, promotions, account status, and disputes. Sensitive actions shall create audit records.
FR-19	Reporting and reconciliation	P1	Authorized administrators shall view basic booking, payment, refund, discount, and settlement reports and compare internal records with provider records.
FR-20	Account and data handling	P0	The system shall support account status changes and apply documented data retention/deletion procedures, subject to legal and financial recordkeeping obligations.

5. Business Rules and Workflows

5.1 Booking Workflow

1. Customer signs in, selects a service, enters job details, and chooses an eligible worker.
2. System shows the service price, platform-funded discount, customer payable amount, cancellation terms, and applicable fees before confirmation.
3. Booking is created in a pending state. Payment timing and worker acceptance order must match the selected provider's supported flow.
4. Worker accepts or rejects the request. Rejection or timeout after payment must trigger the defined reversal/refund workflow.
5. Worker performs the service; completion is recorded by the permitted actor(s).
6. The system records completion, payment status, platform fees/discount expense, and provider settlement status.
7. Eligible customers may submit a review or raise a dispute.

5.2 Booking Statuses

Recommended states: REQUESTED, AWAITING_PAYMENT, PAID, CONFIRMED, IN_PROGRESS, COMPLETION_PENDING, COMPLETED, CANCELLED, REFUND_PENDING, REFUNDED, DISPUTED. The implementation shall define allowed transitions, role permissions, timeout behavior, and compensating actions. Not every booking must pass through every state.

5.3 Pricing and Discounts

Each booking shall preserve the original agreed price, platform-funded discount, customer payable amount, platform fees, and worker earning/settlement basis. Example: a ₹500 service with a ₹50 platform-funded discount results in ₹450 payable by the customer. Worker earnings follow the agreed commercial model; the platform discount must not silently reduce them.

5.4 Financial Integrity

- Store monetary values as integer paise or an exact decimal type, with currency explicitly recorded.
- Do not use floating-point arithmetic for money.
- Do not silently edit completed financial history; use reversals or compensating entries.
- Keep promotional credits separate from payment, refund, and settlement records.
- Use database transactions for related booking and financial updates.

6. Data Requirements

The following are logical entities. Final columns, constraints, indexes, and relations shall be defined in the Prisma schema.

Entity	Purpose / key data
User	Identity, verified phone, role, account status, timestamps.
CustomerProfile	Customer-specific details and profile preferences.
WorkerProfile	Worker details, biography, service area, availability, verification status.
Address	User-linked service or contact address; access restricted to authorized parties.
ServiceCategory	Service name, description, active status.
WorkerSkill	Worker-to-category mapping, experience, pricing.
VerificationRecord	Provider/reference, verification status, reviewer, timestamps; minimize sensitive data.
Booking	Customer, worker, service, schedule, address reference, status, price snapshot, discount, fees.
BookingStatusHistory	Old/new status, actor, timestamp, reason.
Payment	Booking, provider order/payment IDs, amount, currency, status, timestamps.
Settlement	Worker, booking, provider payout reference, amount, status, timestamps.
Refund	Payment, provider refund reference, amount, reason, status.
Promotion	Code, eligibility, discount rules, caps, validity, budget, usage limits.
PromotionRedemption	Promotion, customer, booking, discount amount, status.
LedgerEntry	Auditable financial event type, amount, currency, reference, timestamp.
Review	Booking, customer, worker, rating, optional comment.
Dispute	Booking, reporting party, reason, status, resolution.
Notification	Recipient, event/channel, delivery status, timestamp.
AuditLog	Actor, action, affected entity, timestamp, and minimal necessary metadata.
ProviderEvent	Unique provider event ID, event type, processing status, received timestamp.

6.1 Data Constraints

- Use foreign keys, indexes, and uniqueness constraints where appropriate (including provider event IDs).
- Enforce one eligible review per booking and prevent duplicate promotion redemption where policy requires it.
- Avoid storing raw card data, plaintext OTPs, or unnecessary identity documents.
- Restrict verification records and addresses by role and business need.
- Back up the database and test restoration periodically.

7. External Interfaces

Interface	Requirement
Web frontend	Responsive Next.js interface for customers, workers, and administrators.
REST API	Express API over HTTPS; validate all input and return consistent error responses.
PostgreSQL / Prisma	Relational persistence, migrations, transactions, constraints, and indexed queries.
OTP/SMS provider	Send login OTPs and essential service notifications; enforce rate limits and cost controls.
Payment provider	Payment initiation, verified webhooks, refunds, worker onboarding/settlement, and reconciliation. Marketplace capability must be confirmed before selection.
Identity verification	Approved provider or controlled manual workflow; implementation depends on the permitted legal route.
Object storage (if used)	Store permitted uploads outside the application server; apply private access and retention controls.

8. Non-Functional Requirements

ID	Area	Requirement
NFR-01	Performance	For the controlled pilot, target 95% of normal API reads under 1 second and booking/payment API responses under 2 seconds, excluding external provider latency. Validate with realistic load tests.
NFR-02	Availability	Target 99% monthly application availability during the pilot, excluding planned maintenance and external provider outages.
NFR-03	Scalability	Use stateless API design where practical, indexed queries, pagination, and externalized uploads so the app can move to a larger or multi-instance deployment.
NFR-04	Security	Use HTTPS, secure session handling, input validation, least-privilege access, rate limiting, secret management, and dependency updates.
NFR-05	Privacy	Collect only necessary personal data, explain purposes, limit access, define retention periods, and implement applicable privacy obligations.
NFR-06	Reliability	Make payment callbacks idempotent, handle retries safely, and avoid duplicate bookings, charges, refunds, or settlements.
NFR-07	Auditability	Record booking status changes and privileged or financial operations with actor and timestamp.
NFR-08	Backup and recovery	Configure automated backups where

		available; document recovery steps and test restoration before launch.
NFR-09	Maintainability	Use version-controlled code, environment-specific configuration, Prisma migrations, and automated tests for critical flows.
NFR-10	Compatibility	Support current major versions of common mobile and desktop browsers with a responsive layout.
NFR-11	Monitoring	Track CPU, RAM, disk, API errors, latency, database connections, payment failures, and webhook processing.

9. Security, Privacy, and Compliance Dependencies

These items require review and confirmation before live operations; this SRS is not legal advice.

- Payment and settlement: select a provider that expressly supports the intended marketplace model. Confirm merchant-of-record responsibilities, worker onboarding, settlement timing, refunds, disputes, reconciliation, and any applicable regulatory requirements.
- Aadhaar/identity: do not implement proprietary Aadhaar authentication. Establish the lawful, consent-based verification route and permitted data handling with an appropriately authorized provider or documented manual process.
- Privacy and consumer terms: prepare terms of service, privacy notice, pricing/discount disclosures, cancellation/refund policy, grievance handling, and data retention/deletion procedures.
- Tax and accounting: confirm applicable treatment of commissions, booking fees, promotional expenses, worker payouts, invoices, and GST with a qualified Indian adviser.
- Operational safety: define no-show, misconduct, safety complaint, fraud, suspension, and escalation procedures.
- Production launch is blocked until payment-provider workflow, identity-verification route, and customer-facing policies are approved.

10. MVP Scope and Acceptance Criteria

10.1 Included

- Customer and worker mobile OTP registration, profiles, roles, and addresses.
- Worker verification submission and administrator review.
- Service catalogue, worker search, skills, pricing, and availability.
- Booking request, acceptance/rejection, status history, and cancellation.
- Provider payment integration, signed/idempotent webhooks, refunds, and settlement status tracking.
- Platform-funded promotions, transaction ledger, and customer/worker financial history.
- Essential notifications, basic reviews, disputes, and administrator tools.
- Operational reports, backups, monitoring, and launch testing.

10.2 Excluded

- Proprietary real-money wallet, user deposits, peer-to-peer transfers, or cash withdrawal.
- Unapproved direct Aadhaar authentication or unnecessary retention of full Aadhaar data.
- Multi-city launch, advanced automatic matching, subscriptions, bulk hiring, and automated voice-call infrastructure.

10.3 MVP Acceptance Criteria

8. A customer can verify a mobile number, find an eligible worker, view the complete price, and submit a booking.
9. A worker cannot accept jobs before required verification and payout onboarding are approved.

10. Booking transitions reject unauthorized or invalid changes and preserve status history.
11. A valid promotion applies once within configured eligibility and budget limits; invalid or exhausted promotions are rejected.
12. Payment status changes only after validated provider evidence; duplicate webhook delivery does not duplicate financial effects.
13. Cancellation and refund outcomes are recorded and visible to authorized parties.
14. Worker settlement status is linked to provider references and can be reconciled.
15. Users cannot access another user's private profile, booking, address, or financial records.
16. Administrators can review verification, manage promotions, and resolve disputes with audit records.
17. Backup restoration, critical booking/payment tests, and a realistic load test pass before public launch.

11. Assumptions, Constraints, and Open Decisions

11.1 Assumptions and Constraints

- Initial market is one local area in India; the first service category is not yet finalized.
- Frontend: Next.js. Backend: Node.js with Express. Database: PostgreSQL. ORM: Prisma.
- The ledger records transactions and promotional credits; it does not hold transferable or withdrawable user funds.
- Worker settlement is handled by a payment provider-supported workflow.

11.2 Decisions Required Before Production

Decision	Why it matters
First service category and launch locality	Defines service fields, worker matching, operational policies, and initial demand validation.
Payment provider and fund flow	Determines onboarding, payment timing, settlement, refunds, reconciliation, and compliance obligations.
Worker earnings, platform commission, and any booking fee	Defines price calculations, ledger entries, worker settlement basis, and accounting.
Identity verification provider/process	Defines lawful collection, review steps, data minimization, and worker activation.
Cancellation, no-show, dispute, and refund policy	Determines allowed booking transitions and financial reversals.
SMS provider and notification budget	Affects OTP delivery, operating costs, and rate limits.

Appendix A. Suggested Initial API Surface

Area	Endpoints
Authentication	POST /auth/send-otp; POST /auth/verify-otp
Workers	POST /workers/profile; POST /workers/verification; GET /workers
Bookings	POST /bookings; GET /bookings/:id; PATCH /bookings/:id/status
Promotions	POST /promotions/validate
Payments	POST /payments/create-order; POST /payments/webhook; GET /transactions
Settlements	GET /workers/me/settlements
Reviews / disputes	POST /bookings/:id/reviews; POST /bookings/:id/disputes

All endpoints are proposed. Each must authenticate the caller, authorize access to the target resource, validate inputs, apply rate limits where appropriate, and return documented error codes.

Appendix B. Implementation Note

This document defines requirements, not a guarantee of capacity or legal compliance. Performance targets and infrastructure sizing must be verified through load tests. Payment, identity verification, tax, and privacy arrangements must be confirmed with the selected providers and qualified advisers.